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COLLEGE OF ENGINEERING & TECHNOLOGY SHEGAON

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Department of Electronics and Telecommunication Engineering

Question Bank (Computer Architecture)

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Unit 1:-Structure of Computers

- • • 1) Explain the various functional units of computer in detail (6M)
- • 2) Explain the basic operational concept with the help of connection diagram between Processor & Memory (7M)
- • • 3) Explain memory locations w.r.t. i)Byte addressability ii) Big Endian & Little Endian assignment (7M)
- • • 4) Explain with help of diagram how basic input & output operations are performed .(6M)
- • 5) List the operations performed by various instructions of a computer.Explain in detail instruction execution & instruction sequencing (7M)
- 6) What do you mean by addressing mode ?Explain various types of addressing modes with example .(7M)

Unit 2:-Processing Unit

- 1) Draw & explain data processing part of simple floating point arithmetic unit (7M)
- 2) Give the implementation of four logical instructions AND,OR,EX-OR,& NOT with proper diagram (7M)
- 3) Explain IEEE 754 standard 32 bit floating point number format with the help of an example (7M)
- 4) Construct 16 bit ALU from 4 bit ALU slices (AMD2901) & Explain .(6M)
- 5) Draw & explain structure of a basic fixed point ALU .(6M)
- 6) Explain a simple accumulator based CPU along with flowchart showing sequence of Microinstructions involved (7M)
- 7) Draw & explain n-bit two's complement Adder,subtractor (7M)

Unit 3:-Control Unit

- 1) Explain with help of diagram design of hardwired control unit using sequence counter method (7M)
- 2) List the various ways of program control transfer within a single program & explain the powerful mechanism for transferring control between programs (7M)
- 3) How microprogrammed control unit is superior than hardwired control unit ?Explain microprogrammed unit based on Wilkes design ?(7M)
- 4) What do you mean by 'Parallelism in microinstructions' ?Explain (7M)
- 5) Explain in detail the design of hardwired control unit using delay element method .(7M)
- 6) What is difference between hardwired control and Microprogrammed control ?Explain (7M)
 - i)Horizontal microcode compaction ii) Indirectly data dependant microoperations iii)Weekly data dependant microprogram (7M)

Unit 4:-Memory Unit

- 1) What is Virtual Memory ?What are the reasons for use of virtual memory ?Explain two level & three level memory hierarchies .(7M)
- 2) Explain Paging & Segmentation (6M)
- 3) What is Cache Memory ? Explain caches w.r.t.a) Cache design ii) Address mapping iii)updating main memory (7M)
- 4) The following page address stream is representative for a particular two-level virtual memory system that use demand paging & LRU replacement
3 4 2 6 4 7 1 3 2 6 3 5 1 2 3 Plot a graph of page Hit ratio as a function of Main memory page capacity n for $1 \leq n \leq 8$.Assume that main memory is empty initially.(7M)
- 5) Explain non-preemptive memory allocation techniques in details (6M)
- 6) What are the various replacement policies ?Compare various replacement policies for the given page of address stream considering capacity of cache memory is of 3 page.(7M)

Time (t)	1	2	3	4	5	6	7	8	9	10	11	12
Stream	2	3	2	1	5	2	4	5	3	2	5	2

- 7) Explain the following points w.r.t.memory device characteristics i)cost ii) Access time iii)Access Mode iv) Alterability v) Permanence of storage vi) Cycle time & data transfer rate vii)Physical Characteristics (7M)

Unit 5:-I/O Organization

- 1) What do you mean by interrupt ? Explain transfer control through the use of Interrupt (6M)
- 2) Give the implementation of interrupt priority using INTR & INTA lines.Explain interrupt priority scheme based on it (7M)
- 3) Explain in details operations & various features of PCI bus (7M)
- 4) Draw & explain a general 8 bit parallel interface circuit (6M)
- 5) What is USB ? What are the objectives ? Explain the USB tree structure & its operations (7M)
- 6) What is Bus Arbitration ? Explain Centralized & Distributed Arbitration scheme (7M)
- 7) What is DMA ?Explain data transfer in a computer system with DMA techniques (6M)
- 8) What is need of DMA controller ? & draw block diagram of DMA controller ? Justify?(7M)

Unit 6:-Parallel Processing

- 1) What is multiprocessor?Explain its types detail with the help of neat figure (8M)
- • • 2) Discuss Flynn's classification of processor parallelism (6M)
- • 3) Explain organization of a pipelined CPU with five segment instruction pipeline (6M)
- 4) In a certain scientific computation ,it is necessary to perform the arithmetic operations (A_i+B_i) , (C_i+D_i) with a stream of numbers.Specify the pipeline configuration to carry out this task.List the content of all registers in pipeline for $i=1$ to 6 (8M)
- 5) Explain in detail structural classification of parallel computers (6M)
- 6) Explain the organization of a pipelined CPU with five segment instruction pipeline (6M)
- • 7) What is pipeline processors ?Explain two segments instructions pipeline & four segments floating point adder pipeline (6M)
- • 8) Draw & explain pipelined microprogram control unit (6M)